

Inverse Functions and Inverse Trigonometry

Quick Reference

Definition of an Inverse Function

A function f has an inverse function only when it is **one-to-one**: distinct inputs must produce distinct outputs. Equivalently, its graph must pass the horizontal line test. The inverse reverses every input-output pair:

$$f(a) = b \iff f^{-1}(b) = a.$$

The notation f^{-1} means the inverse function, not the reciprocal $1/f$.

If f is one-to-one, then

$$\text{Dom}(f^{-1}) = \text{Range}(f), \quad \text{Range}(f^{-1}) = \text{Dom}(f).$$

Algebraically, inverse functions undo one another:

$$f^{-1}(f(x)) = x \quad \text{and} \quad f(f^{-1}(x)) = x,$$

whenever x belongs to the appropriate domain. Graphically, the graphs of f and f^{-1} are reflections across the line $y = x$. Their points have reversed coordinates:

$$(a, b) \text{ on } y = f(x) \iff (b, a) \text{ on } y = f^{-1}(x).$$

The Six Inverse Trigonometric Functions

Trig functions must be restricted to one-to-one intervals before inverses can be defined. This sheet uses the following standard principal-value conventions:

Function	Domain	Range (principal values)
$\sin^{-1} x$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1} x$	$(-\infty, \infty)$	$(-\frac{\pi}{2}, \frac{\pi}{2})$
$\csc^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$[-\frac{\pi}{2}, 0) \cup (0, \frac{\pi}{2}]$
$\sec^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$[0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$
$\cot^{-1} x$	$(-\infty, \infty)$	$(0, \pi)$

For an allowed input, a trig function outside its matching inverse cancels directly, such as

$$\sin(\sin^{-1} x) = x \quad (-1 \leq x \leq 1).$$

In the reverse order, cancellation occurs only when the angle already lies in the inverse function's principal range. For example,

$$\sin^{-1}(\sin \theta) = \theta \quad \text{only when} \quad -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}.$$

For different functions, interpret the inverse-trig expression as an angle in its principal range and construct a right triangle, keeping its quadrant in mind.

Inverse Trigonometry and Inverse Functions

30-Minute Practice

Name: _____

Date: _____

Instructions

Complete all 12 questions in 30 minutes without a calculator. Show enough work to justify each answer. Use radians, give exact values, and state all domain restrictions. Do not assume that trig and inverse-trig functions cancel unless the relevant principal-range condition is satisfied.

1. Evaluate each expression exactly.

$$(a) \sec\left(\sec^{-1}\left(-\frac{5}{2}\right)\right) \quad (b) \csc\left(\csc^{-1}\left(\frac{7}{3}\right)\right) \quad (c) \cot\left(\cot^{-1}(-\sqrt{3})\right)$$

2. Evaluate each expression exactly.

$$(a) \sec^{-1}\left(\sec \frac{5\pi}{3}\right) \quad (b) \csc^{-1}\left(\csc \frac{7\pi}{6}\right) \quad (c) \cot^{-1}\left(\cot \frac{5\pi}{4}\right)$$

3. Evaluate exactly:

$$\sec\left(\tan^{-1}\left(-\frac{5}{12}\right)\right).$$

4. Evaluate exactly:

$$\tan\left(\cos^{-1}\left(-\frac{5}{13}\right)\right).$$

5. Evaluate exactly:

$$\sin\left(\cot^{-1}\left(-\frac{7}{24}\right)\right).$$

6. Solve over the real numbers:

$$\sin(2 \tan^{-1} x) = \frac{24}{25}.$$

7. Let

$$f(x) = \sqrt{2x-6} - 1.$$

State the domain and range of f , find $f^{-1}(x)$, and state the domain and range of f^{-1} .

8. Let

$$g(x) = \frac{3x+2}{x-4}.$$

Find $g^{-1}(x)$. State the excluded value in the domain and range of both g and g^{-1} .

9. Let

$$h(x) = (x - 3)^2 - 4, \quad x \geq 3.$$

Find $h^{-1}(x)$ and state the domain and range of both functions. Explain why the restriction $x \geq 3$ is necessary.

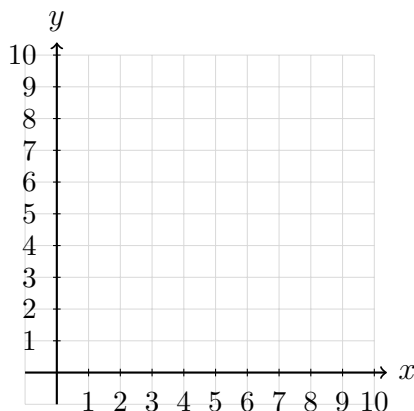
10. The graph of a one-to-one function p contains the points

$$(-3, 2), \quad (0, -1), \quad (4, 5), \quad (7, 6).$$

- (a) List the corresponding points on the graph of p^{-1} .
- (b) State the geometric transformation that maps the graph of p to the graph of p^{-1} .
- (c) If $p(4) = 5$, interpret $p^{-1}(5) = 4$ in terms of inputs and outputs.

11. Consider $q(x) = |x - 2| + 1$.

- (a) Explain graphically why q has no inverse function on its full domain.
- (b) Restrict the domain to $x \geq 2$ and find the resulting inverse.
- (c) On the grid, sketch the restricted function, its inverse, and the line $y = x$.



12. Let

$$r(x) = 2(x + 1)^3 - 5.$$

Find $r^{-1}(x)$. Then verify algebraically that

$$r(r^{-1}(x)) = x \quad \text{and} \quad r^{-1}(r(x)) = x.$$

State what these two identities mean about the input-output actions of inverse functions.